Rafael Ode Brino

Brest, France | German & Brazilian citizenship
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Six-month ENSTA final-year research internship starting in March 2027

Education

ENSTA — Institut Polytechnique de Paris Brest, France
MSc in Engineering — Observation Systems & Artificial Intelligence; expected September 2027 *2025–2027*
Selective French engineering school (*grande école*); concurrent studies with UFRGS.

- Relevant coursework: Machine Learning, Deep Learning, Numerical Optimization & Markov Models, Digital Image Processing, Software Design, and Decision Theory & Estimation.

Federal University of Rio Grande do Sul (UFRGS) Porto Alegre, Brazil
BSc in Electrical Engineering; expected August 2028 *2023–2028*

- Relevant coursework: Data Structures, Algorithms & Programming, Object-Oriented Programming, Probability & Statistics, Linear Algebra, and Numerical Calculus.

Research Experience

NATO Science & Technology Organization (STO) La Spezia, Italy
Research Intern — Centre for Maritime Research and Experimentation (CMRE) *May–September 2026*

Article 1: Hierarchical Bayesian Modeling of Detection Success *(first author; in preparation)*

- Implemented **hierarchical Bayesian probit models** in Python for object-detection success, combining partial pooling and MCMC with analytical marginalization for predictions in unseen data-collection groups.
- Achieved **group-averaged ROC AUC 0.734** on 21 held-out groups; nested validation and paired group bootstraps supported ranking gains over a prevalence baseline, while richer features added no stable benefit.

Article 2: Modeling Detector Performance Under Image Degradation *(first author; in preparation)*

- Built a Python/PyTorch pipeline for YOLO training and evaluation on Slurm-managed HPC; ran paired tests across **64 conditions** of resolution, blur, and noise, delivered reusable code, and presented findings at CMRE.
- Calibrated a nonlinear model of detector performance, reducing prediction **RMSE by 48% (0.237 to 0.122)** versus reusing the source coefficients unchanged; evaluated with fully nested selection on held-out noise levels.

Selected Projects

FX Risk Forecasting | *Python, C++, SQL, DuckDB* 2026–Present

- Built a Python/SQL (DuckDB) pipeline for ECB exchange rates and implemented HMM, Student-t GARCH, and independent-mixture models to test whether volatility regimes improve forecasts of currency-return distributions.
- Implemented a **C++ HMM filter** with Python bindings and numerical parity checks; built rolling out-of-sample evaluation with training-cutoff checks and paired block-bootstrap comparisons.

Augmented Reality for Surgical Training | *Python, C++, C#, Unity* 2025–2026

- Built a computer-vision pipeline for HRIACT – French Armed Forces hospital, mapping **physical scalpel movements** from Python/C++ to a virtual instrument in a Unity/C# surgical-training simulation.

3D Mapping with Detection and SLAM | *Python, YOLO* 2024–2025

- Co-developed an EKF-SLAM pipeline fusing stereo visual odometry, YOLO detections, and GPS for 3D mapping and trajectory estimation; coauthored the **published CBA 2026 preprint** (paper; English abstract).

Technical Skills

Programming: Python, C++, SQL, Bash, MATLAB, C#

ML & Data: PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, pandas, OpenCV, DuckDB

Modeling & Inference: Bayesian inference, MCMC, HMMs, EM, nonlinear optimization, Kalman filtering

Development & HPC: Git (GitHub/GitLab), Linux, pytest, pybind11, SSH, Slurm, Neovim

Research & Simulation: JupyterLab, Unity, Simulink, Matplotlib, LaTeX

Languages

Portuguese: Native | English: Fluent | French: Advanced | German: Elementary